

Permutations - solving a case study of your choice[← Back](#)

Practice Assignment • 30 min

**Due** Jul 10, 11:59 PM IST**Your grade: 100%**Your latest: **100%** • Your highest: **100%**

To pass you need at least 80%. We keep your highest score.

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1. In the Permutations video, we asked you to consider the case of finding the right actors for the roles. Try to write and solve a case of your choice.

1 / 1 point

For instance, if I have a bag of 10 sweets and three children to share them with, how many ways can the sweets be divided among the children? Remember to consider extreme cases (e.g. where one child gets all 10 sweets and the other two do not get any sweets).

 $12!/2!/10!=66$ **Feedback****Model answer**

This is a classic problem of distributing n indistinguishable objects into k distinguishable bins, which can be solved using the ‘stars and bars’ theorem.

The stars and bars theorem states that the number of ways to distribute n indistinguishable objects into k distinguishable bins is given by the binomial coefficient:

$$\binom{n+k-1}{k-1}$$

In this problem, sweets are our objects, so n is 10, and children are bins, so k is 3. Replace these values to find the answer.